

From chapter 1



Chapter 5

Risk Management

BAIT3153 Software and
Project Management (SPM)

A high-angle photograph of a group of seven diverse professionals (four men and three women) gathered around a large, dark wooden conference table. They are engaged in a collaborative meeting. Several laptops and tablets are open on the table, displaying various data and charts. One man in a white shirt is pointing at a tablet, while others look on attentively. The setting is a bright, modern office with large windows in the background.

Chapter 5

Risk Management

5.1 Introduction

- What is risk?
- What is risk mgmt?
 - **Reactive** risk strategies Vs **proactive** risk strategies

5.2 Risk Management Processes

5.3 Risk Management **Planning**

5.4 Risk **Identification**

5.5 Risk **Analysis, Projection**

5.6 Risk **Refinement**

5.7 Risk **Mitigation, Monitoring, & Management (RMMM)**

5.1 Introduction

What is Risk?

- **An uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives**
- **A risk is "something" that you would prefer not to have happen**
(project risk may threaten *completion time*, lead to *cost overrun*, result in *low product quality*, etc)

What is risk management?

Is the **art** and **science** of identifying, assigning, and responding to risk throughout the life of a **project** in the best interests of meeting project objectives

Examples of project risks

Risks	Category	Probability	Impact	Risk Score
High staff turnover	ST	60%	3	1.8
Funding will be lost	ST	30%	3	0.9

Customer keep changing requirements

5.1 Introduction

❑ What is Risk?

An uncertain event or condition that, if it occurs, has a positive or negative effect on a project's objectives

❑ What is Risk Mgmt?

Is the art and science of identifying, assigning, and responding to risk throughout the life of a project

Did you identify **potential risks** before starting your FYP?

e.g.

FYP Risks	Category	Probability	Impact (1 – 4) (1=negligible 4 = catastrophic)	RMMM
Large project scope		50 %	3	Mitigation ... Monitoring ...
Hardware failure (laptop need to be sent for repairs)		1 %	3	Mitigation ... Monitoring ...

- ✓ Involves understanding potential problems that might occur on the project and how they might impede project success.
- ✓ The goal of project risk management is to **minimize potential risks** while maximizing potential opportunities.
- ✓ It is often **overlooked in projects**, but it can help to improve project success by helping select good projects, determining project scope, and developing realistic

5.1 Introduction

Reactive Risk Strategies (RRS):

- ❑ Take actions to deal with the risks **after** problems have arisen.

❑ Aka Fire-fighting

E.g Provide training to team members **after** seeing them struggling with unfamiliar tools.

Proactive Risk Strategies (PRS):

- ✓ Identify potential risks, assess the probability and impact **before** any technical work begins.
- ✓ Develop a response plan to address the risk in a controlled and effective manner.

(means preparing in advance what to do if identified risk materialises)

E.g Provide training to team members on unfamiliar tools **before** project starts.

5.2 Risk Management Processes

Risk planning:

Deciding how to approach and plan the risk management activities for the project

Risk identification:

Determining which risks are likely to affect a project and documenting their characteristics

Qualitative **Risk analysis:**

Characterizing and analysing risks and prioritizing their effects on project objectives

Quantitative risk analysis:

Measuring the probability and consequences of risks

Risk response planning:

Taking steps to enhance opportunities and reduce threats to meeting project objectives

Risk monitoring and control:

Monitoring known risks, identifying new risks, reducing risks, and evaluating the effectiveness of risk reduction

5.3 Risk Management Planning

A Risk Management Plan should cover the followings areas:

- ❑ **What** is the specific risk? *What could go wrong?*
- ❑ **Why** it is important to consider the risk *If it happened, how bad would it be?*
- ❑ **How** is the risk going to be mitigated? *How to reduce the probability & lessen the damage?*
- ❑ **How much efforts** are required to mitigate the risk?
- ❑ **Who is responsible** for implementing the risk management plan?

<u>E.g</u> Risk	Why...	Mitigation	Effort	Who is responsible
1. Laptop failure	Could delay FYP progress	<u>Backup</u> FYP(docs & codes) to the cloud	1h daily	Student
2. Academic deadline clashes	Test & assignments preparation may reduce coding time.	<u>Create</u> a project schedule with <u>buffer</u> in days/weeks.	2h planning	Student

5.3 Risk Management Planning

A Risk Management Plan should cover the followings areas:

- ❑ **What** is the specific risk?
(What could go wrong?)
- ❑ **Why** it is important to take or not to take this risk
(How bad would it be if it happened?)
- ❑ **How** is the risk going to be mitigated?
(How to reduce the probability & lessen the damage?)
- ❑ **How much efforts** are required to mitigate the risk?
- ❑ **Who is responsible** for implementing the risk management plan?

Also useful to have:

- ~~Contingency plans - predefined actions to take if an identified risk becomes real (e.g. if UI expert A is not available when needed then the contingency plan is to use UI expert B)~~
- Reserved funds to mitigate risk if it does occur
(e.g. funds to pay an outside consultant to **train inexperienced staffs** when inexperienced staffs are underperformed)

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- ❑ **Why** it is important to take or not to take this risk
- ❑ **How** the risk is going to be mitigated?
- ❑ **How** much efforts are required to mitigate the risk?
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Also useful to have:

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5.4 Risk Identification

Identify risks that could threaten the ***project completion time, cost, product quality , etc***

— Risk identification is a ~~systematic attempt to specify threats to the project plan in order to avoid them when possible and controlling them when necessary.~~

— 2 distinct types of risks:

i) Generic risks

- ~~potential threat to every software project (e.g., schedule risk)~~

ii) Product-specific risks

- ~~can only be identified by those with a clear understanding of the technology, the people, and the environment that is specific to the project.~~
- ~~e.g., in medical systems, accuracy is a risk~~

5.4 Risk Identification - **Techniques**

project team members get together to discuss possible risks ...

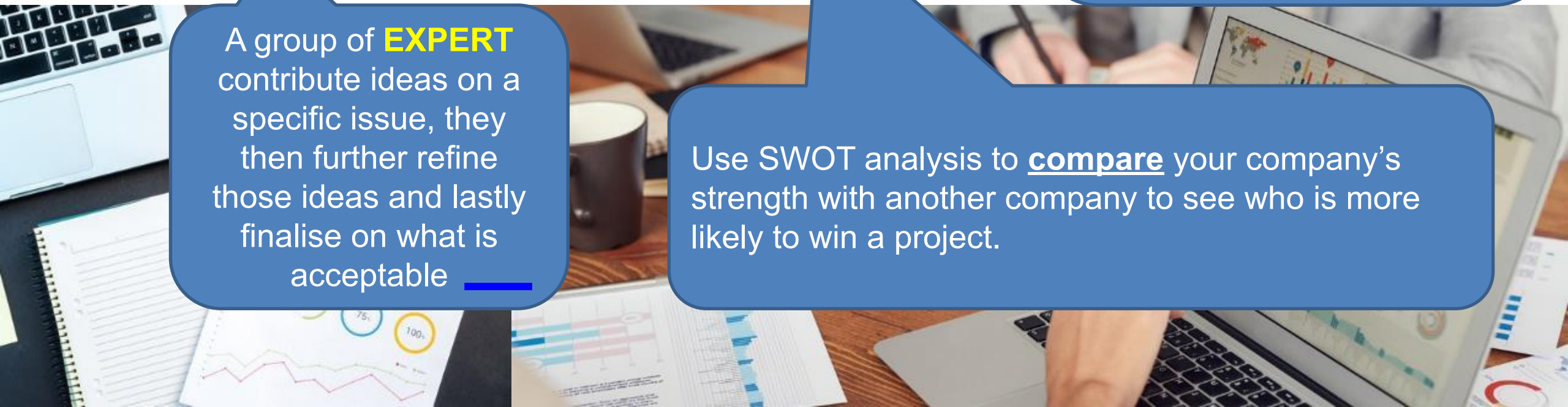
- **Brainstorming**
- The **Delphi** technique

A group of **EXPERT** contribute ideas on a specific issue, they then further refine those ideas and lastly finalise on what is acceptable —

- **Interviewing**
- **SWOT** analysis

- **Ask people** who have used particular hardware, software & OS in his previous project (e.g. uipath)
- **Ask people** who have worked with a particular customer before you (e.g. customer A)

Use SWOT analysis to compare your company's strength with another company to see who is more likely to win a project.



5.4 Risk Identification - Assessing Overall Project Risk

Use Questions to identify project risks (e.g.s. below)

- ▶ 1. Can the organization **afford** to undertake the project?
- 2. Is this project the **best way** to use the company's financial resources?
- 3. Does the organization have the **people** with the right **skills** to complete the project successfully?
- ▶ 4. Does senior management **support** the project?
- 5. Will **hardware**, **software** and **networks** function properly?
- 6. Will the user workflows or existing business procedures remain unchanged after project implementation?

If any ANSWER to these questions is "NO": **mitigation, monitoring & management** steps should be implemented.

Rank the risks by their relative **importance to the success of a project**

5.5 Risk Management Processes

Risk planning:

Deciding how to approach and plan the risk management activities for the project

Risk identification:

Determining which risks are likely to affect a project and documenting their characteristics

Risk analysis:

Qualitative

Characterizing and analysing risks and prioritizing their effects on project objectives

Quantitative

Estimate the probability and impact of each risks

Risk response planning:

Take steps to enhance opportunities and reduce threats to meeting project objectives

Risk monitoring and control:

Monitor known risks, identifying new risks, reducing risks, and evaluating the effectiveness of risk reduction

5.5 Risk Projection – Risk Table

- The risk table provides PM with a simple technique for risk projection.
- Steps to **develop a risk table**:
 - Column 1 – risk identified
 - Column 2 – **category** of each risk →
 - Column 3 – **probability** for each risk
 - Column 4 – **impact** of each risk (1 to 4 or 1 to 10)
 - Column 5 – **RMMM** (risk mitigation, monitoring, management)

5.5 Risk Projection –Category of Risk

CATEGORY of RISK	DESCRIPTION: Risks associated with
☐ Technology to be built (TE)	the complexity of the system to be built and the “newness” of the technology that is packaged by the system. (E.g.1 using a new language in a project) (E.g.2 compatibility issues when completed software needs to integrate with existing legacy systems)
☐ Staff size and experience (ST)	The overall technical and project experience of the software engineers who will do the work. (e.g.1 operating with a smaller team than required) (e.g.2 team members lack experience in critical areas like UI design and security)
☐ Customer characteristics (CU)	the sophistication of the customer and the developer’s ability to communicate with the customer in a timely manner. (e.g.1 customer has unrealistic expectations) (e.g.2 customer keep changing requirements)
☐ Product size (PS)	the overall size of the software to be built or modified.
☐ Business impact (BU)	constraints imposed by management or the marketplace.
☐ Process definition (PR)	the degree to which the software process has been defined and is followed by the development organization.

5.5 Risk Projection (Risk Estimation)

- Risk projection/estimation attempts to **rate each risk in 2 ways:**
 - the likelihood or **probability** that the risk is real
 - the consequences or **impact associated with the risk**
(How bad would it be if it happened?)
- 4 risk projection activities:
 1. Establish a scale that reflects the perceived likelihood of a risk
 2. Estimate the impact of the risk
 3. Identify the consequences of the risk
 4. **Ensure accuracy to avoid misunderstandings.**

5.5 Risk Projection – Risk Impact Category & Scale

Risk impact assessment is carried out based on this Risk Impact Categories. The categories for each of the 4 risk components – performance, support, cost and schedule – are averaged to determine an overall impact value.

Note:

- (1) The potential consequence of undetected software errors or faults.
- (2) The potential consequence if the desired outcome is not achieved.

Components Category	Performance		Support	Cost	Schedule
Catastrophic 4	1	Failure to meet the requirement would result in mission failure		Failure results in increased costs and schedule delays with expected values in excess of \$500K	
	2	Significant degradation to nonachievement of technical performance	Nonresponsive or unsupportable software	Significant financial shortages, budget overrun likely	Unachievable IOC
Critical 3	1	Failure to meet the requirement would degrade system performance to a point where mission success is questionable		Failure results in operational delays and/or increased costs with expected value of \$100K to \$500K	
	2	Some reduction in technical performance	Minor delays in software modifications	Some shortage of financial resources, possible overruns	Possible slippage in IOC
Marginal 2	1	Failure to meet the requirement would result in degradation of secondary mission		Costs, impacts, and/or recoverable schedule slips with expected value of \$1K to \$100K	
	2	Minimal to small reduction in technical performance	Responsive software support	Sufficient financial resources	Realistic, achievable schedule
Negligible 1	1	Failure to meet the requirement would create inconvenience or nonoperational impact		Error results in minor cost and/or schedule impact with expected value of less than \$1K	
	2	No reduction in technical performance	Easily supportable software	Possible budget underrun	Early achievable IOC

5.5 Risk Projection – Risk Table Example

Risks	Category	Probability	Impact	RMMM
Less reuse than planned	PS	70%	3	
End-users resist system	BU	40%	2	
Funding will be lost	CU	40%	4	
Lack of training on tools	DE	80%	2	
High staff turnover	ST	60%	3	
Staff inexperienced	ST	30%	3	
Size estimate may be significantly low	PS	60%	3	
Larger number of users than planned	PS	30%	2	
Delivery deadline will be tightened	BU	50%	3	
Customer will change requirements	PS	80%	2	
Technology will not meet expectations	TE	30%	4	

Impact:
 1 – negligible,
 2 – marginal,
 3 – critical,
 4 – catastrophic

Risk
**Mitigation,
 Monitoring &
 Management
 Plan**

5.5 Risk Projection – Risk Prioritization

- Once the first 4 columns of the risk table have been completed, sort the RISK table by probability and by impact.
- The PM studies the sorted table and defines a cutoff line.
- Risks above the cutoff line to be given further attention. Risks below the line are to be re-evaluated
- **Pay attention to:**
 - High-impact risks with moderate to high probability
 - low-impact risks but with high probability
- The column labeled RMMM contains a pointer to a risk mitigation, monitoring, and management plan or a collection of risk information sheets

OR

Calculate the **Risk Score** for each risk. **Pay attention to** those risks with **high score**.

Risk Score = Probability * Impact

Sort the risk table by Risk Score in descending order.

Example

Calculate Risk Score:

Risks	Category	Probability	Impact	Risk score
Less reuse than planned	PS	70%	2	$0.7 \times 2 = 1.4$
End users resist system	BU	40%	3	$0.4 \times 3 = 1.2$
Funding will be lost	CU	40%	1	$0.4 \times 1 = 0.4$
Lack of training tools	DE	80%	3	$0.8 \times 3 = 2.4$
High staff turnover	ST	60%	2	$0.6 \times 2 = 1.2$

1. Sort risk table by “Risk score” in descending order

Example

Calculate Risk Score:

Risks	Category	Probability	Impact	Risk score
Lack of training tools	DE	80%	3	$0.8 \times 3 = 2.4$
Less reuse than planned	PS	70%	2	$0.7 \times 2 = 1.4$
End users resist system	BU	40%	3	$0.4 \times 3 = 1.2$
High staff turnover	ST	60%	2	$0.6 \times 2 = 1.2$
Funding will be lost	CU	40%	1	$0.4 \times 1 = 0.4$

1. Sort risk table by “Risk score” in descending order

(Sorted risk table by risk score in descending order)

2. Set cutoff line

5.5 Risk Projection – Assessing Risk Exposure (RE)

(Means calculate the amount of losses when a particular risk becomes real)

If costs are associated with a risk, Halstead's **risk exposure (RE)** metric can be calculated and added to the risk table using:

$$RE = P \times C$$

P: probability of occurrence for a risk

C: cost to the project should the risk occur

EXAMPLE

60 reusable software components were planned. Only 70% (42 components) can be reused, 18 components would have to be developed from scratch. The average component is 100 LOC and the cost for each LOC is \$14.00.

Risk identified: Only 70% out of 60 software components can be reused. The remaining 30% (18 components) have to be custom developed.

Risk probability: 80%

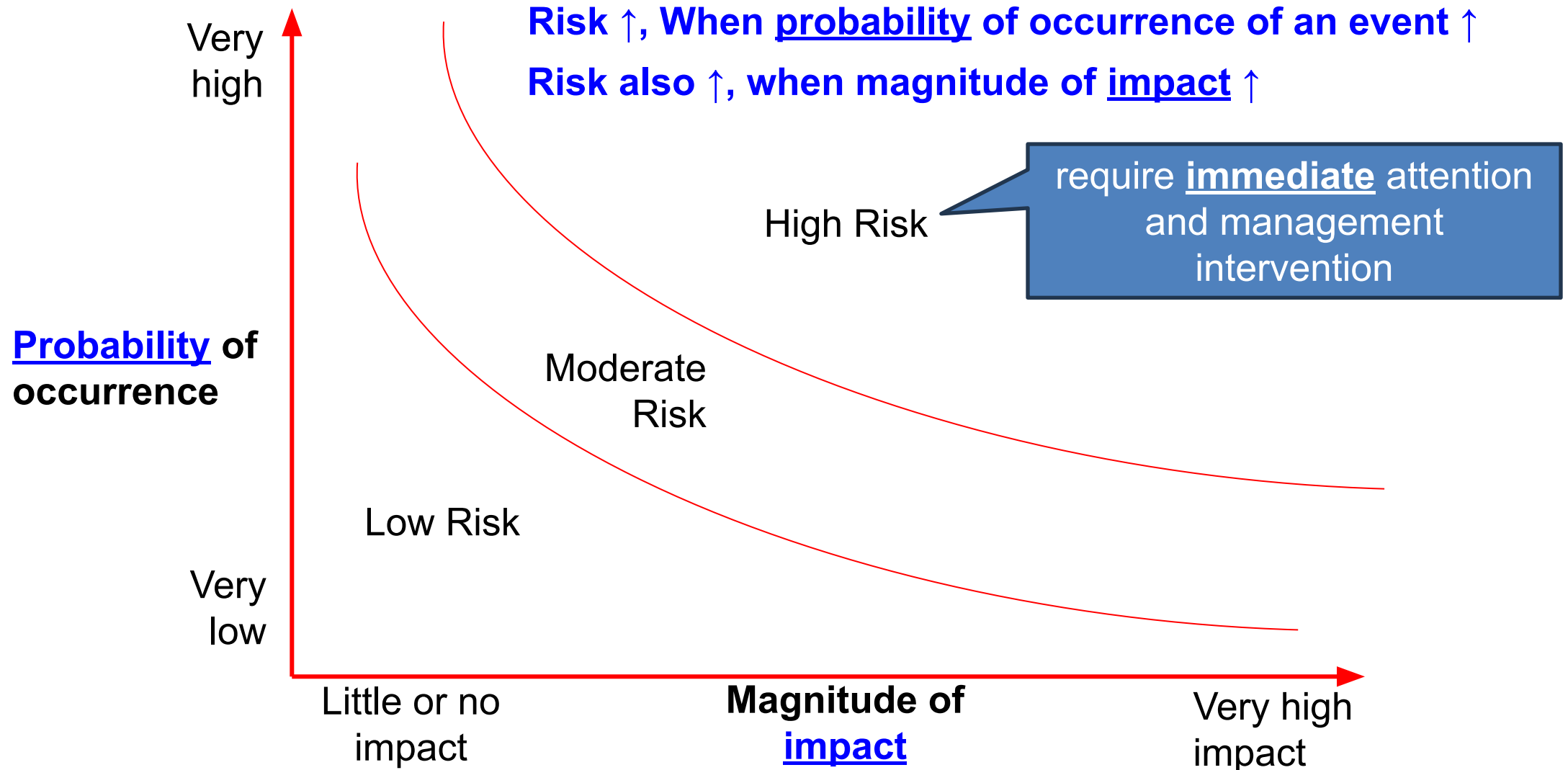
The cost to develop the 18 components = $18 \times 100 \times 14 = \$25,200$.

Risk exposure: $RE = P \times C$
 $= 0.80 \times 25,200$
 $= \$20,160$

amount of losses if the mentioned risk becomes real

5.5 Risk Projection – Risk & Management Concern

Risk has 2 primary components:



5.5 Risk Projection – Consequences

During risk assessment, we consider 3 factors affect the consequences that are likely if a risk does occur:

Nature

- What problems are likely to occur
- e.g. a poorly defined external interface to customer hardware (a technical risk) will preclude early design and testing and will likely lead to system integration problems late in a project.

Scope

How serious is the problem (severity) with its overall distribution (how much of the project will be affected or how many customers are harmed?)

Timing

When and for how long the impact will be felt

5.6 Risk Refinement

- During early stages of project planning, a risk may be stated quite generally.
- Later as more is learned about the project and the risk, it may be possible to refine the risk into a set of more detailed risks to make it easier to mitigate, monitor and manage.
 - The risk may be stated using the condition-transition-consequence (CTC) format:

Given that <condition> then there is concern that (possibly)<consequence>

Example

Initial Risk (<i>general</i>)	Refined Risk (<i>specific</i>)
<i>"UI may not meet expectations"</i>	Poor UI design may increase help line support from 5/hr to 20/hr
<i>"Project scope might expand"</i>	<i>Client change requests may increase project scope by 50%.</i>
<i>"High staff turnover"</i>	When a Sr. Developer quit, 3 more people might quit

5.7 Risk Mitigation, Monitoring, & Management (RMMM)

Risks	Category	Probability	Impact	RMMM
High staff turnover	ST	60%	3	Mitigation ... Monitoring ... Management ...
Staff inexperienced	ST	30%	3	
Funding will be lost	CU	40%	4	
Lack of training on tools	DE	80%	2	

An effective risk management strategy must consider 3 issues:

a. Risk mitigation

To develop options and actions to reduce threats to project objectives

b. Risk monitoring & control

A project tracking activity with 3 objectives:

- (i) assess whether predicted risks do, in fact occur;
- (ii) ensure that risk avoidance steps defined for the risk are being properly applied; and
- (iii) carry out the risk management plans as risks occur

The main outputs of risk monitoring and control are corrective action, project change requests, and updates to other plans .

c. Risk management and contingency planning

Take actions when the risk has become a reality.

Risk Information Sheet

Risk ID: P03-1-37

Date: 15/2/2020

Probability: 70%

Impact: Critical

Description:

Based on past history and management intuition, high staff turnover will have a critical impact on project cost and schedule.

Mitigation:

= Reduce the probability of the risk occurring (make it less likely to happen).

= Reduce the impact if the risk does occur (make the impact less severe).

- Meet with
- Mitigate th
- Review and adjust salary packages and benefits to be competitive before the project start. This is to avoid staff from leaving for better pay elsewhere.
- Disperse work product standards and establish mechanisms to ensure that all models and documents are developed in a timely manner.
- Conduct peer reviews of all work.
- Assign backup staff member for every critical technology.

E.g. Assign **Jack** to work closely with **Jim** to manage the project's network infrastructure. When **Jim** is unavailable due to resignation then **Jack** can ensure service continuity

Monitoring:

= A risk tracking activity (PM may collect information to do risk analysis)

- Monitor the general attitude of team members based on project pressures.
- Check the degree to which the team has jelled.
- Assess the interpersonal relationships among team members.
- Check potential problems with compensation and benefits.
- Monitor the availability of jobs within the company and outside.

E.g. Monitor the overall economy of the country and IT job mkt (check _____)

Management:

= Take actions when the risk has becomes a reality

If any staff announce that they will be leaving when the project is under way, following the mitigation strategy will ensure that:

- Use backup staff that has been planned in the mitigation stage
- E.g. use **Jack** to replace those who has left the company
- Knowledge has been dispersed across the team

In Ask the individual who is leaving to spend the last week to transfer knowledge to remaining team members

enabling newcomers who must be added to the team to get up to speed. The individuals who are leaving will be asked to stop all work and spend their last weeks in knowledge transfer mode. This might include video-based knowledge capture, the development of commentary documents or Wikis, and/or meeting with other team members who will remain on the project.

5.7 Risk Mitigation, Monitoring, & Management (RMMM)

“A **risk** is something that you would prefer not to have happen ”

Risk management strategies

- **Risk avoidance:** Take the necessary actions to avoid the risk altogether.
- **Risk reduction:** Take precautions to reduce the probability of the risk.
- **Risk mitigation:** Take the necessary actions to reduce the impact of the risk when it occurs.
- **Risk transfer:** Transfer the risk to another person or organization.

- **Risk acceptance:** The “**do nothing**” option. In the risk prioritization process, it would have been decided for some risks to be ignored in order to concentrate on the more likely or damaging risks. For some risks the **costs of action > the damage** inflicted in order to reduce the probability of a risk from happening.

Summary

- 5.1 Introduction
- 5.2 Risk Management Processes
- 5.3 Risk Management Planning
- 5.4 Risk Identification
- 5.5 Risk Projection
- 5.6 Risk Refinement
- 5.7 Risk Mitigation, Monitoring, & Management (RMMM)

